AN INTELLIGENT AGENT FOR FAILURE ANALYSIS USING HUMAN-IN-THE-LOOP LEARNING

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Contents

1. Broad Area of Work

Integrating Artificial Intelligence into DevOps transforms traditional manual **root cause analysis (RCA)** of operational pipelines into intelligent Cognitive AIOps workflows. This dissertation investigates how advanced AI techniques can be embedded into automated pipelines and infrastructure workflows by leveraging **Large Language Models (LLMs)** and **Agentic Retrieval-Augmented Generation (RAG)** to generate actionable resolutions for failures. The proposed AI agent combines **stochastic reasoning** for probable cause prediction with **function-calling** capabilities for executing real-time validation checks, while maintaining **Human-in-the-Loop (HITL)** collaboration to ensure safety, governance, and continuous learning. This approach enhances software pipeline reliability, operational resilience, and intelligent automation across modern enterprise systems.

The current research project focuses on the following application areas:

- **AIOps** – Automates IT operations, monitoring, and incident management.
- **DevOps** – Enhances software delivery speed, pipeline reliability and improves operational stability.
- **Large language Models (LLMs) & Generative AI** – Enables intelligent log analysis and decision support.
- **Autonomous Agents** – Performs automated diagnostics and remediation tasks.
- **Agentic RAG** – Retrieves relevant contextual knowledge for accurate root cause analysis.
- **Human-in-the-Loop (HITL) Systems** – Ensures safe AI decisions through human oversight and feedback.

2. Background

Modern software engineering has rapidly evolved toward high-velocity DevOps ecosystems, where complex operational pipelines, infrastructure workflows, deployment systems, and cloud environments drive continuous software delivery and business operations. As organizations increasingly rely on automated pipelines and workflows, failures across these systems have become a critical operational challenge. Traditional root cause analysis (RCA) methods are often manual, time-consuming, and dependent on human expertise, leading to slower incident response, prolonged downtime, and reduced operational resilience. While Site Reliability Engineering (SRE) practices improve reliability, existing approaches often lack adaptive intelligence, proactive reasoning, and scalable automation.

Recent advancements in Artificial Intelligence for IT Operations (AIOps), Large Language Models (LLMs), Autonomous Agents, and Retrieval-Augmented Generation (RAG) offer transformative potential for modern operational management. By combining real-time log analysis, infrastructure health validation, organizational knowledge repositories, historical incidents, and external technical resources, intelligent systems can automate diagnosis, remediation planning, and workflow optimization across diverse operational environments. This dissertation explores the integration of cognitive AIOps, agentic systems, and Human-in-the-Loop (HITL) learning to create a self-evolving framework capable of diagnosing failures, optimizing remediation, and improving resilience across software pipelines and operational workflows.

3. Objectives

- To analyze existing DevOps, AIOps, and SRE tools and design a Cognitive Agentic framework and architecture for intelligent operational failure management.
- To implement an Agentic Retrieval-Augmented Generation (RAG) system that integrates internal and external knowledge sources for context-aware root cause analysis and resolution planning.
- To develop tool-augmented autonomous diagnostic agents capable of performing real-time infrastructure validation, health checks, and operational analysis.
- To integrate Human-in-the-Loop (HITL) governance mechanisms for safe decision-making and continuous system learning.
- To optimize operational reliability by evaluating system performance through metrics such as MTTR reduction, RCA accuracy, and remediation effectiveness.
- To build a scalable self-evolving operational intelligence model for adaptive enterprise automation.

4. Scope of Work

The scope includes intelligent root cause analysis, real-time validation through tool integration, remediation planning, and Human-in-the-Loop governance for safe decision-making. It emphasizes improving operational reliability, automation, and resilience while delivering a scalable self-evolving model for modern DevOps, infrastructure workflows, and Site Reliability Engineering environments.

5. Plan of Work

Phases	**Start Date-End Date**	**Work to be done**
Dissertation Outline & Design	10 May 2026 – 18 May 2026	Literature Review, Architectural Analysis and Design of a Cognitive Agent
Development	25 May 2026 – 15 July 2026	Development Activity
Testing	16 July 2026 – 24 July 2026	Software Testing, User Evaluation & Conclusion
Dissertation Review	25 July 2026 - 2 Aug 2026	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	4 Aug 2026 - 10 Aug 2026	Final Review and submission of Dissertation

6. Literature References

The preliminary literature review focuses on emerging advancements in intelligent operational automation, autonomous reliability systems, and Human-AI collaborative frameworks. The following references represent key scholarly contributions and state-of-the-art research that support the conceptualization and design.

1. Li, X., Wang, S., Zeng, S., Wu, Y., & Yang, Y. **(2024)**. A survey on LLM-based multi-agent systems: Workflow, infrastructure, and challenges. *Vicinagearth, 1*(1), Article 9. Available at: Springer Link

2. Wang, L., Ma, C., Feng, Y., Zhang, Z., Yang, H., Liang, H., et al. **(2024)**. A survey on large language model based autonomous agents. *Frontiers of Computer Science, 18*(6), 186345. Available at: Springer Link

3. Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., Wiest, O., & Zhang, X. **(2024)**. Large language model based multi-agents: A survey of progress and challenges. *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, 8048–8057. Available at: IJCAI Proceedings

4. He, J., Treude, C., & Lo, D. **(2024)**. LLM-based multi-agent systems for software engineering: Literature review, vision and the road ahead. *arXiv preprint arXiv:2404.04834*. Available at: arXiv

5. Zhang, H., Liu, Y., Chen, P., & Zhao, J. **(2025)**. AIOpsLab: A holistic framework to evaluate AI agents for enabling autonomous clouds. *arXiv preprint arXiv:2501.06706*. Available at: arXiv

6. Chen, J., Rao, S., Patel, V., et al. **(2025)**. STRATUS: A multi-agent system for autonomous reliability engineering of modern clouds. *OpenReview Conference Paper*. Available at: OpenReview

7. Demir, K., & Yilmaz, A. **(2025)**. From playbooks to autonomous operations: How LLMs are redefining incident management in site reliability engineering. *International Journal of Computational and Experimental Science and Engineering*. Available at: IJCESEN Journal

8. Fernando, M., & Rajan, P. **(2024)**. LLM-based autonomous remediation for DevSecOps pipelines. *Eastasouth Journal of Information System and Computer Science*. Available at: Eastasouth Journal

9. Kumar, D., Lee, J., & Park, S. **(2024)**. A taxonomy of AgentOps for enabling observability of foundation model-based agents. *ResearchGate Preprint*. Available at: ResearchGate

10. Sharma, R., Gupta, N., & Chen, Y. **(2024)**. Building AI agents for autonomous clouds: Challenges and design principles. *ResearchGate Preprint*. Available at: ResearchGate

7. Particulars of the Supervisor and Examiner

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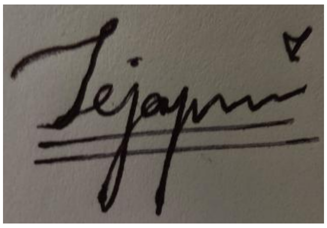
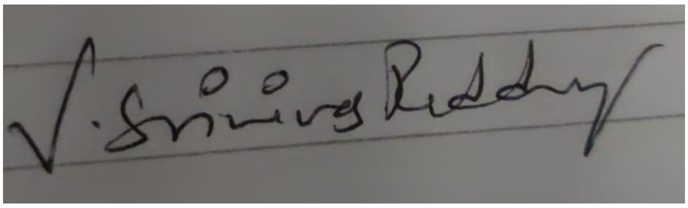
8. Remarks of the Supervisor

The proposed project offers significant practical value by enhancing organizational efficiency, operational decision-making, and collaborative problem-solving within modern technology environments. By introducing intelligent automation into complex workflows, the research can support engineering teams in reducing manual effort, accelerating issue resolution, and improving productivity across software and infrastructure operations. The system's ability to assist teams with faster diagnostics and actionable insights can strengthen enterprise reliability while enabling more streamlined and effective resource management.

This dissertation also demonstrates strong potential to support organizations in adopting next-generation operational practices through scalable AI-driven solutions. Its focus on intelligent assistance, team collaboration, and adaptive workflow optimization makes it highly relevant for real-world enterprise applications. The project is well-suited for master-level research and presents promising opportunities for practical deployment, innovation, and future advancements in organizational technology management. Based on the student's technical capabilities, research direction, and project design, the dissertation is approved as a valuable and promising contribution to the field.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